

User: Alice Mar 18

Who organized my longest outdoor activity in March?

Question → **LLM Agent** → Query → **MemCatalog**

Answer ← **LLM Agent** ← Retrieved memory ← **MemCatalog**

How to use?

How to store?

For Accuracy — Query Planning

(a) Single-fact retrieval

Who organized the hike?

target
hike-1

Ground → Verify

subject	predicate	object
hike-1	organizer	Ava

Select

value
Ava

(b) Independent composition

How many March activities were outdoors?

activity	setting
hike-1	outdoor
pottery-1	indoor
ride-1	outdoor

Evaluate *outdoor?*

activity	yes?
hike-1	1
pottery-1	0
ride-1	1

Combine *1 + 0 + 1*

count
2

(c) Dependent composition

Who organized my longest outdoor activity in March?

outdoor	duration
hike-1	4 h
ride-1	3 h

Compare *4 h > 3 h*

longest
hike-1

Bind

target	property	value
hike-1	organizer	Ava

For Cost — Cost-aware Execution

work	state	LLM calls
reuse known facts	verified	0
hike duration	missing	estimated
ride duration	missing	estimated

Priorities and cost estimates

Allocate and execute

Within the query budget

Re-estimate after each action

1. Raw Sessions

Mar 2	User: "Ava arranged our 4-hour outdoor hike at Pine Trail."
Mar 9	User: "Ben booked my 2-hour indoor pottery class."
Mar 16	User: "Cara arranged our 3-hour outdoor ride at Lake Park."
Mar 18	User: Who organized my longest outdoor activity in March? Agent: Ava organized the 4-hour hike.

Extract missing facts

Verify against User spans

2. First-order Memory

activity	property	value	source
hike-1	setting	outdoor	Mar 2
pottery-1	setting	indoor	Mar 9
ride-1	setting	outdoor	Mar 16
hike-1	organizer	Ava	Mar 2
pottery-1	organizer	Ben	Mar 9
ride-1	organizer	Cara	Mar 16
hike-1	duration	4 h	Mar 2
ride-1	duration	3 h	Mar 16

Use / Update

3. Second-order Memory

Reported organizer use (Mar 17)

organizer fact	retrieved	used
hike: Ava	yes	yes
pottery: Ben	yes	no
ride: Cara	yes	yes

Use / Update

action	cost	outcome
acquire duration	logged	verified